Vansh

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Education

B.Tech in Computer Science Engineering, Vellore Institute of Technology *Sep 2023 – Expected Aug 2027* | ***8.81 CGPA***
Class XII, Floradale School, Ganganagar, Rajasthan ***84.8%***
Class X, D.A.V. Public School, Suratgarh, Rajasthan ***95.33%***

Technical Skills

Languages: Java, Python, C++, JavaScript, TypeScript
Web Technologies & Libraries: React, Node.js, Express.js, Next.js, HTML/CSS, Tailwind CSS, Shadcn, Socket.io, JWT
Machine Learning & AI: Pandas, NumPy, Large Language Models (LLMs), Prompt Engineering
Databases: NoSQL, MongoDB, MySQL, PostgreSQL, SupaBase, Redis (Upstash), Prisma
Cloud & DevOps: AWS, Docker, CI/CD, GitHub Actions, Vercel, Netlify, Render
Tools & Testing: Git, GitHub, Playwright
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), REST APIs

Experience

Software Development Engineer Intern — Digi Marking Infotech Pvt Ltd *Jun 2026 – Aug 2026 | Hybrid*

- Validated **UI components** and **30+ REST API payloads** across **3 client projects** to resolve edge cases, authoring **11+ bug reports** and system documentation to standardize workflows.
- Refactored **10+ complex monolithic frontend files** by isolating state logic into **25+ reusable UI components**, reducing code redundancy and significantly improving **debugging speed**.
- Automated **40+ end-to-end tests** with **Playwright** covering **authentication workflows** and routing, then delivered the suite to DevOps for **CI integration** to prevent staging regressions.

Web Development Intern — KL LAB *May 2025 – Jul 2025 | Remote*

- Engineered secure and scalable **REST APIs** with **Node.js** and **Express.js**, adding **JWT authentication** that protected **20+ endpoints** and cut unauthorized access attempts by **30%**.
- Modeled **MongoDB schemas** with **Mongoose**, speeding up data retrieval for user, blog, and comment workflows by **40%** and cutting query latency to **under 50ms**.
- Deployed backend systems to **cloud infrastructure**, collaborating with a 4-person frontend team to achieve seamless **full-stack integration** and maintain **99.9% uptime** post-launch.

Technical Team Core Member — AWS Club, VIT Bhopal *Oct 2024 – Jul 2025 | Bhopal, MP*

- Orchestrated **5+ technical workshops** on **AWS fundamentals** for **100+ members**, driving a **20% increase** in club engagement and leading team-based **cloud architecture projects**.

Projects

AI-Powered ATS Resume Screener | **GitHub** | **Live Demo** ***Next.js, React, Shadcn, Redis, Prisma, PostgreSQL, TypeScript, LLMs***

- Architected an **enterprise ATS simulation platform** that evaluates resumes concurrently against **Legacy** (Taleo, Workday) and **Modern** (Greenhouse, Eightfold) systems, using distinct parser strictness and **semantic evaluation rules**.
- Reduced API latency and token consumption by **50%** via **LLM prompt consolidation** and implemented **SHA-256 normalised response caching** via **Upstash Redis**, reducing repeated scan latency by **97%** — from **37 seconds to under 1 second**.
- Built a **high-availability inference pipeline** cascading from **Gemini 2.5 Flash** to **Groq** (Llama 3.3 70B - Llama 3.1 8B) with **exponential back-off** and a **three-tier sliding window rate limiter** (Redis - PostgreSQL - fail-open), guaranteeing near **100% uptime** during provider rate limits.
- Designed a highly concurrent data architecture featuring a fail-open **Prisma 7 rate-limiting system** and unified **PostgreSQL logging transactions**, deployed behind edge-protected **Clerk v6 middleware**.

Real-Time Network Intrusion Detection System | **GitHub** | **Live Demo** ***React, Express, Node.js, MongoDB, XGBoost, Docker***

- Developed a high-performance **XGBoost binary classifier** pipeline using **Python, NumPy, and Pandas**, trained on **421,959 cleansed network flows** from the CIC-DDoS2019 dataset, achieving **99.85% accuracy** in distinguishing malicious from benign traffic.
- Secured an **Express backend** with **JWT auth**, rate limiting, Helmet headers, Winston logging, and **bcryptjs password hashing**, enabling **sub-second IP blocking** of detected threats via **3 mitigation endpoints**.
- Programmed a **real-time SOC dashboard** with **React and Socket.io**, streaming live threat alerts, packet graphs, **geo-IP enrichment**, and interactive threat maps, plus a **1-second ML prediction** button, with updates in **under 2 seconds**.
- Containerized a 3-service architecture with **3 multi-stage Dockerfiles**, cutting production image sizes by **up to 90%**, and built a **GitHub Actions CI/CD workflow** to automate build validation and **Docker Hub** publishing – **Kubernetes-ready** with rolling updates and health probes.

Certifications

- AWS Cloud Practitioner Essentials - Nasscom